

## ***4.2: Largest Bending Stress in Beams***

- **Let's go back to the simplified version of the flexure formula:**

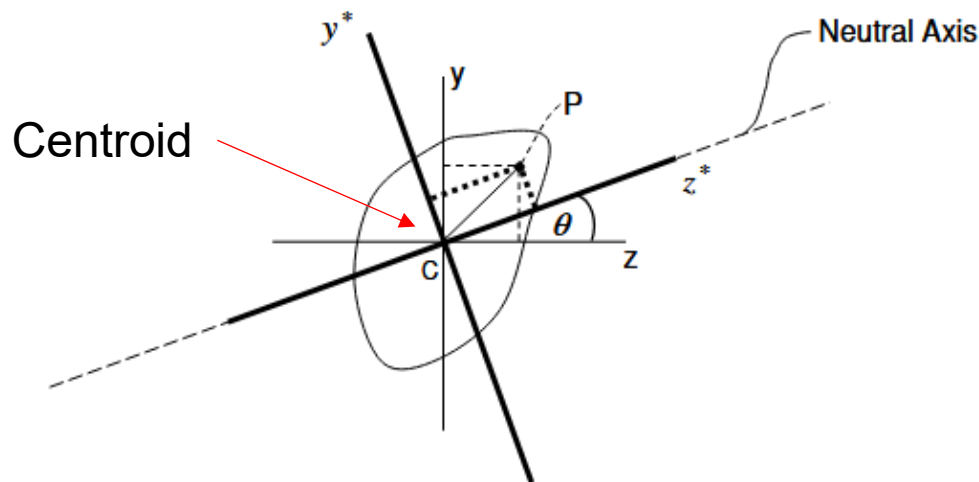
$$\sigma_x = \frac{My}{I}$$

- **We know in this case, the z-axis is always the neutral axis and the largest value of  $\sigma$  on the beam cross-section occurs at the top or bottom (maximum  $y$ )**
- **What about in the general case? We know now how to find the neutral axis; can we make a general statement about the maximum  $\sigma$  location (i.e. critical point) relative to the N.A.?**



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- Below is a general beam cross-section with standard  $y$ - $z$  axes placed at the centroid, as well as  $y^*$ - $z^*$  axes, rotated from  $y$ - $z$  such that the  $z^*$  axis is along the neutral axis:



- The General Flexure Formula is valid for any  $y$ - $z$  coordinate frame with origin at the centroid; applying this to the  $y^*$ - $z^*$  frame yields:

$$\sigma_x = \frac{-y^* (I_{y^*} M_{z^*} + I_{y^* z^*} M_{y^*}) + z^* (I_{y^* z^*} M_{z^*} + I_{z^*} M_{y^*})}{I_{y^*} I_{z^*} - (I_{y^* z^*})^2}$$

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- Since the  $z^*$  axis is the N.A.,  $\sigma_x$  must = 0 when  $y^* = 0$ , so that:

$$0 = \frac{0 + z^* (I_{y^*z^*} M_{z^*} + I_{z^*} M_{y^*})}{I_{y^*} I_{z^*} - (I_{y^*z^*})^2} \Rightarrow (I_{y^*z^*} M_{z^*} + I_{z^*} M_{y^*}) = 0 \Rightarrow M_{y^*} = - \left( \frac{I_{y^*z^*}}{I_{z^*}} \right) M_{z^*}$$

- Inserting this expression for  $M_{y^*}$  back into the  $\sigma_x$  equation yields:

$$\sigma_x = \frac{-y^* \left[ I_{y^*} M_{z^*} + I_{y^*z^*} \left( -\frac{I_{y^*z^*}}{I_{z^*}} M_{z^*} \right) \right] + z^* \left[ I_{y^*z^*} M_{z^*} + I_{z^*} \left( -\frac{I_{y^*z^*}}{I_{z^*}} M_{z^*} \right) \right]}{I_{y^*} I_{z^*} - (I_{y^*z^*})^2}$$

- Grouping terms and simplifying yields:

$$\sigma_x = \frac{-y^* M_{z^*} \left[ I_{y^*} - \frac{(I_{y^*z^*})^2}{I_{z^*}} \right] + z^* [I_{y^*z^*} - I_{y^*z^*}] M_{z^*}}{I_{y^*} I_{z^*} - (I_{y^*z^*})^2} \Rightarrow \sigma_x = \frac{-y^* M_{z^*} [I_{y^*} I_{z^*} - (I_{y^*z^*})^2]}{I_{z^*} [I_{y^*} I_{z^*} - (I_{y^*z^*})^2]} \Rightarrow \sigma_x = \frac{-y^* M_{z^*}}{I_{z^*}}$$



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- We saw recently that we can express  $\sigma_x$  in the simple “ $\frac{My}{I}$ ” fashion when the direction of the applied moment aligns with one of the principal axes (in which case we would align our  $y$  and  $z$  axes with the principal axes as well)
- Here we see that we can also express  $\sigma_x$  in this simplified form if we align the  $z$  axis with the N.A.
  - Although in most cases, we need to use the general form in order to find (i.e. solve the equation for) the N.A.!
- Therefore, we can state that the largest value of  $\sigma$  occurs at maximum  $y$  when the  $z$  axis is aligned with the N.A., i.e. at the greatest perpendicular distance from the N.A.

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- We saw this was the case in the previous section, i.e. the 2 critical points are in fact at the greatest perpendicular distance from the N.A.:

